

#Task 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English"]
hours = [2, 3, 1]

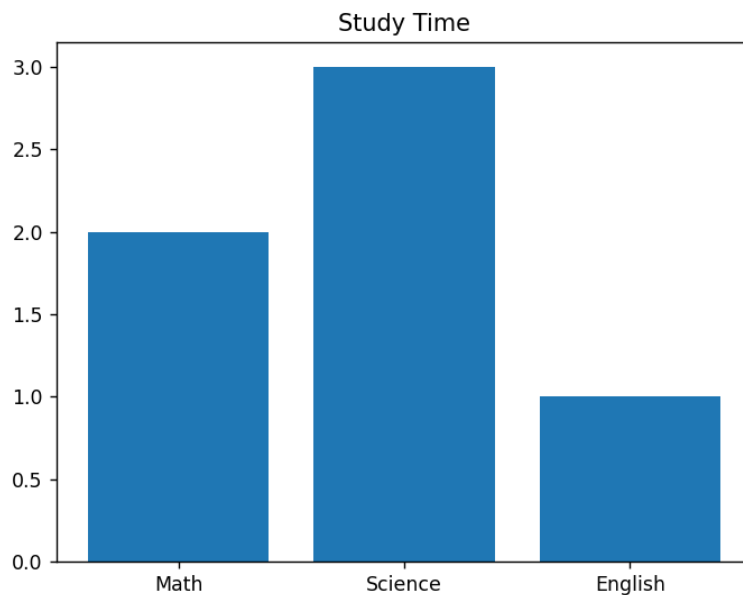
df = pd.DataFrame({"Subject": subjects, "Hours": hours})

total = np.sum(df["Hours"])
avg = np.mean(df["Hours"])

print("Total Study Time:", total)
print("Average Study Time:", avg)

plt.bar(df["Subject"], df["Hours"])
plt.title("Study Time")
plt.show()
```

```
===== RESTART: C:/Users/ki
Total Study Time: 6
Average Study Time: 2.0
```



#Task 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

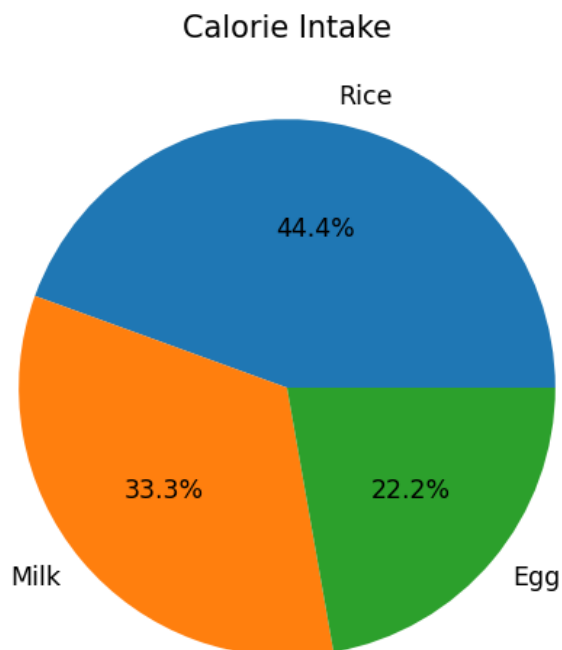
food = ["Rice", "Milk", "Egg"]
calories = [200, 150, 100]

df = pd.DataFrame({"Food": food, "Calories": calories})

total = np.sum(df["Calories"])
print("Total Calories:", total)

plt.pie(df["Calories"], labels=df["Food"], autopct="%1.1f%%")
plt.title("Calorie Intake")
plt.show()
```

```
===== RESTART: C:
Total Calories: 450
```



```
#Task 3
```

```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

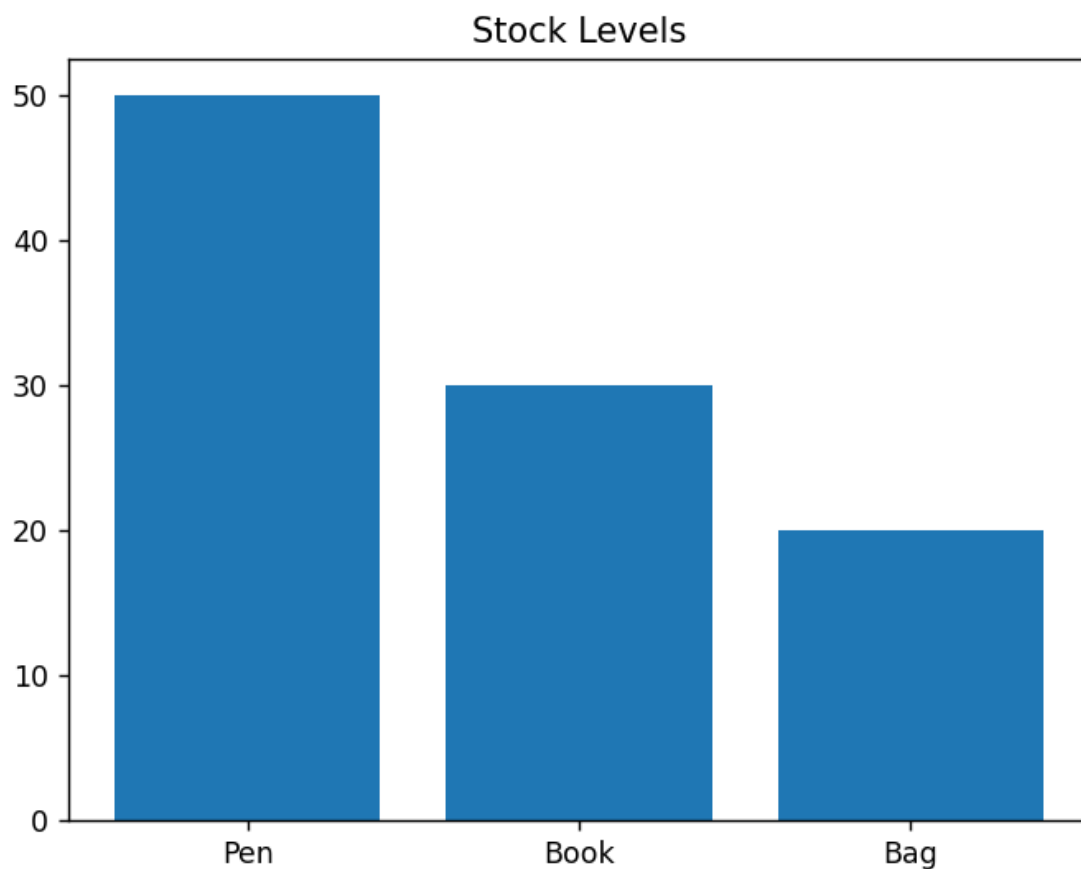
```
products = ["Pen", "Book", "Bag"]  
quantity = [50, 30, 20]
```

```
df = pd.DataFrame({"Product": products, "Quantity": quantity})
```

```
total = np.sum(df["Quantity"])  
print("Total Stock:", total)
```

```
plt.bar(df["Product"], df["Quantity"])  
plt.title("Stock Levels")  
plt.show()
```

```
===== RESTART: C:/Use  
Total Stock: 100
```



#Task 4

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

numbers = [10, 20, 30, 40, 50]

df = pd.DataFrame({"Numbers": numbers})

print("Max:", np.max(df["Numbers"]))
print("Min:", np.min(df["Numbers"]))
print("Average:", np.mean(df["Numbers"]))

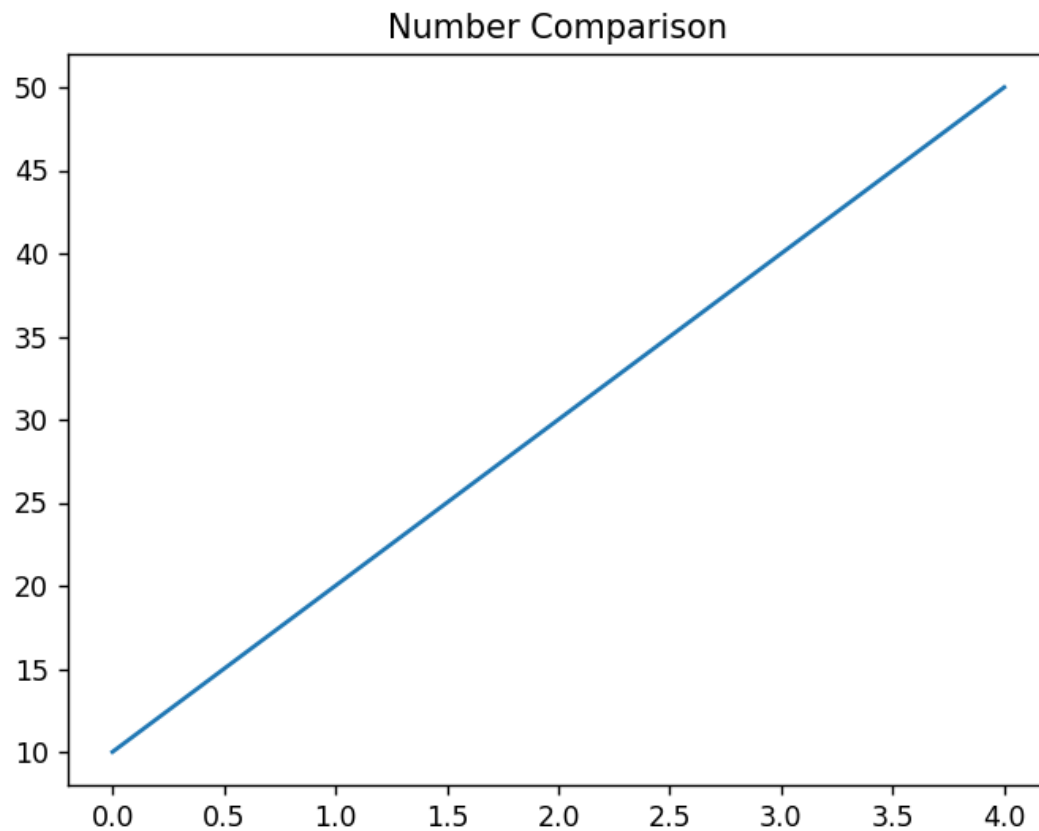
plt.plot(df["Numbers"])
plt.title("Number Comparison")
plt.show()
```

===== RESTART:

Max: 50

Min: 10

Average: 30.0



```
#Task 4
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
players = ["A", "B", "C"]
scores = [80, 60, 90]
```

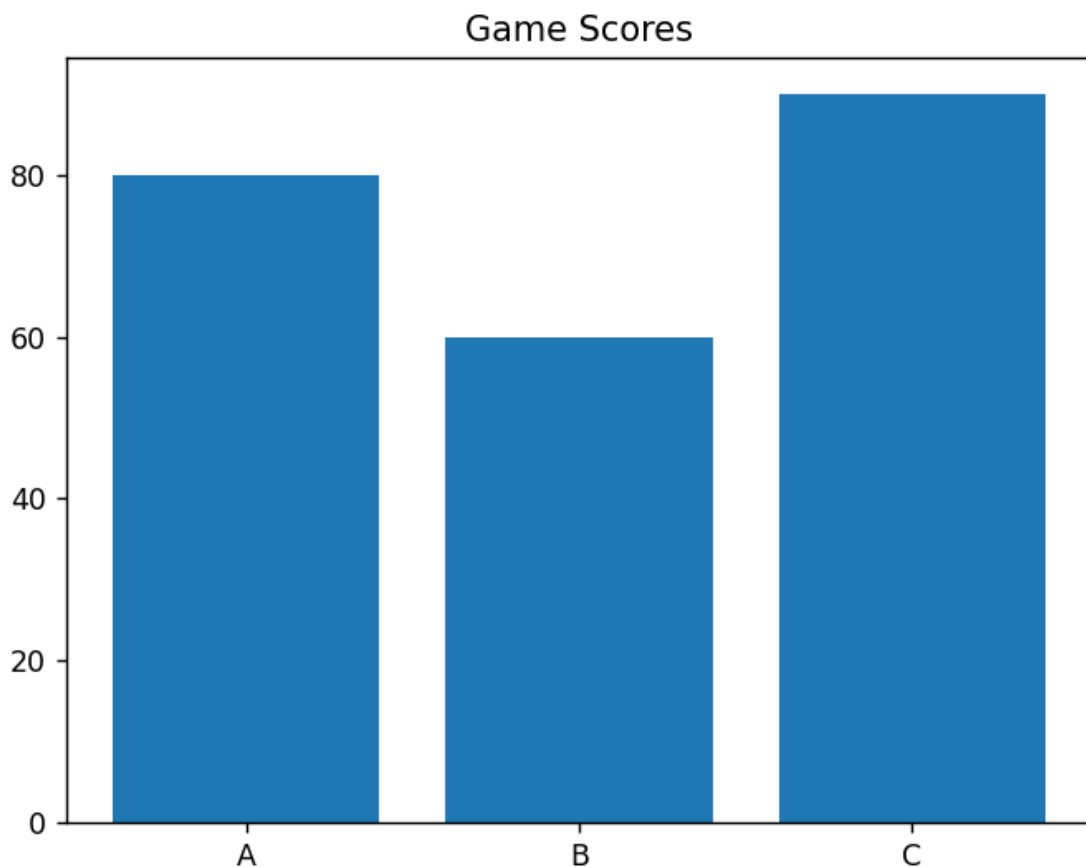
```
df = pd.DataFrame({"Player": players, "Score": scores})
```

```
print("Highest Score:", np.max(df["Score"]))
```

```
plt.bar(df["Player"], df["Score"])
plt.title("Game Scores")
plt.show()
```

```
===== RESTART:
```

```
Highest Score: 90
```



```
#Task 6
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

hours = ["9AM", "10AM", "11AM", "12PM"]
vehicles = [50, 80, 120, 90]

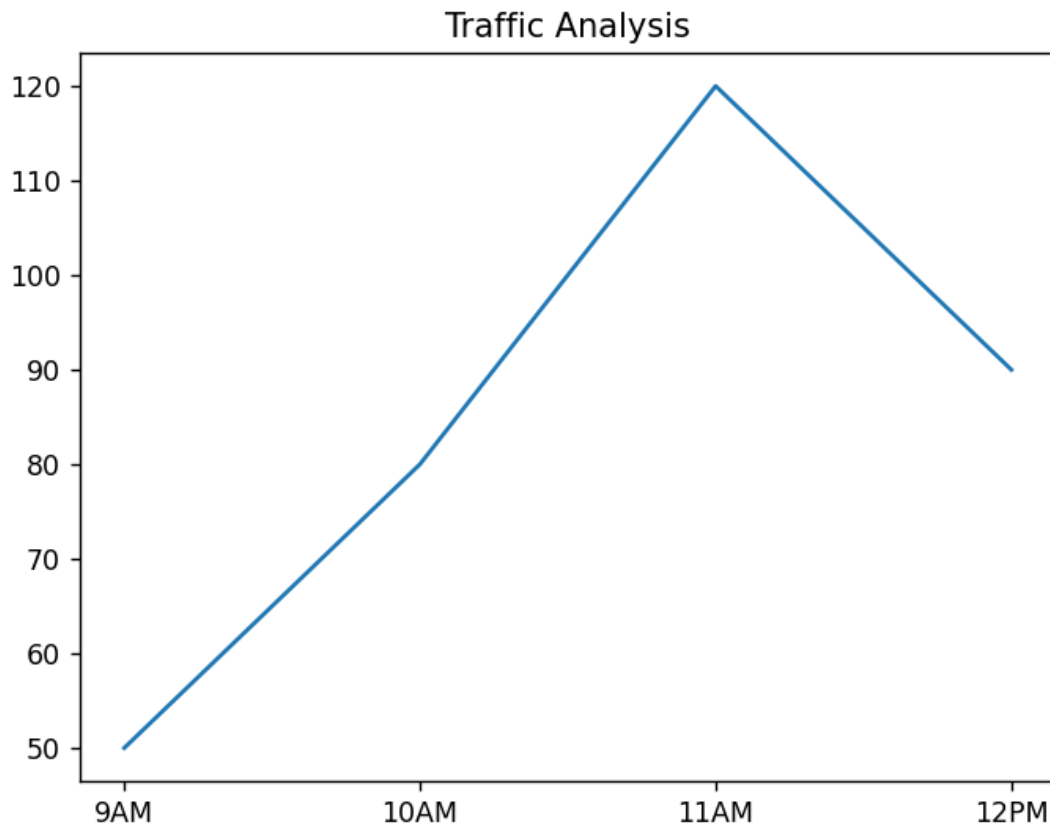
df = pd.DataFrame({"Time": hours, "Vehicles": vehicles})

peak = df.iloc[np.argmax(df["Vehicles"])]
print("Peak Time:", peak["Time"])

plt.plot(df["Time"], df["Vehicles"])
plt.title("Traffic Analysis")
plt.show()
```

```
===== RESTART: C
```

```
Peak Time: 11AM
```



```

# Task 7
TP = 50
FP = 5
FN = 10
TN = 35
Total = TP + FP + FN + TN
Accuracy = (TP + TN) / Total
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1_Score = 2 * (Precision * Recall) / (Precision + Recall)
print("Accuracy =", Accuracy)
print("Precision =", Precision)
print("Recall =", Recall)
print("F1 Score =", F1_Score)
Total = 40 + 5 + 5 + 10 + 30 + 10 + 5 + 5 + 40
Accuracy = (40 + 30 + 40) / Total
Precision_A = 40 / (40 + 10 + 5)
Recall_A = 40 / (40 + 5 + 5)
F1_A = 2 * (Precision_A * Recall_A) / (Precision_A + Recall_A)
Precision_B = 30 / (5 + 30 + 5)
Recall_B = 30 / (10 + 30 + 10)
F1_B = 2 * (Precision_B * Recall_B) / (Precision_B + Recall_B)
Precision_C = 40 / (5 + 10 + 40)
Recall_C = 40 / (5 + 5 + 40)
F1_C = 2 * (Precision_C * Recall_C) / (Precision_C + Recall_C)
print("Accuracy =", Accuracy)
print("F1 Score A =", F1_A)
print("F1 Score B =", F1_B)
print("F1 Score C =", F1_C)
TP = 20
FP = 10
FN = 80
TN = 890
Total = TP + FP + FN + TN
Accuracy = (TP + TN) / Total
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1_Score = 2 * (Precision * Recall) / (Precision + Recall)
print("Accuracy =", Accuracy)
print("Precision =", Precision)
print("Recall =", Recall)
print("F1 Score =", F1_Score)
TP = 45
FP = 15
FN = 5
TN = 35

```

```
Total = TP + FP + FN + TN
Accuracy = (TP + TN) / Total
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1_Score = 2 * (Precision * Recall) / (Precision + Recall)
print("Accuracy =", Accuracy)
print("Precision =", Precision)
print("Recall =", Recall)
print("F1 Score =", F1_Score)
```

```
===== RESTART: C:/Users/krsu/
```

```
Accuracy = 0.85
Precision = 0.9090909090909091
Recall = 0.8333333333333334
F1 Score = 0.8695652173913043
Accuracy = 0.7333333333333333
F1 Score A = 0.761904761904762
F1 Score B = 0.6666666666666665
F1 Score C = 0.761904761904762
Accuracy = 0.91
Precision = 0.6666666666666666
Recall = 0.2
F1 Score = 0.30769230769230765
Accuracy = 0.8
Precision = 0.75
Recall = 0.9
F1 Score = 0.8181818181818182
```

```
>
```